Conformal-Calibrated Rewards for Scientific RLVR: Procedural Regeneration Against Benchmark Contamination

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Abstract

Reinforcement learning from verifiable rewards (RLVR) has become the dominant training signal for frontier reasoning models, but existing verified environments are dominated by symbolic or code-centred tasks. Scientific inverse problems—CT, MRI, compressed sensing, phase retrieval—remain unmeasured despite their continuous, ill-posed, uncertainty-sensitive structure. We release ten RL environments spanning five scientific modalities with two design properties absent from current benchmarks: (i) every reward is split-conformal calibrated to a target $1-\alpha$ coverage, so honest posterior width is rewarded alongside point-estimate quality; and (ii) every measurement is procedurally regenerated per query, making fixed-string contamination mathematically impossible at $\sim 10^{22}$ effective instances per env. On 50 paired (env, model) comparisons across six frontier models, classical baselines significantly outperform every tested LLM on 32 at $p<0.05$ (uncorrected and Bonferroni-corrected), pooled mean $\Delta=+0.199$ (10k paired bootstrap). Top LLMs (Haiku 4.5, Opus 4.7, Sonnet 4.6) reach $0.53-0.56$ cross-env mean, below classical 0.630. An earlier oracle-delegation artefact ($r=0.858$) in the tool-use env was removed; primitive-only reruns across all six models cluster at $0.40-0.55$, against classical OMP at 0.87. Empirical conformal coverage across all ten envs lands at 0.9013 ± 0.0166 against the 0.90 target ($N=200$). Environments are MIT-licensed on the Prime Intellect Hub.

1 Introduction

Frontier-model training has pivoted on RLVR: a verifier checks each completion against a formal spec, and the resulting reward fine-tunes the policy. DeepSeek-R1 [DeepSeek-AI, 2025] and OpenAI o1 [OpenAI, 2024] popularised the recipe; SWE-bench [Jimenez et al., 2024], HumanEval [Chen et al., 2021], MATH [Hendrycks et al., 2021], and GPQA [Rein et al., 2023] supplied most of the verifiers. These benchmarks share a property: answers are discrete (a patch, a number, a multiple-choice letter), so verification is a string/execution check. *Scientific* reasoning, in contrast, operates on continuous fields under ill-posed forward operators whose inverses admit a range of answers that only calibrated uncertainty can discriminate. Classical algorithms in this domain—OMP [Donoho, 2006, Candès and Tao, 2006], FBP [Adler and Öktem, 2018], zero-filled inverse FFT [Fessler, 2020], Gerchberg–Saxton [Gerchberg and Saxton, 1972]—achieve calibrated answers via mathematical guarantees; whether frontier LLMs can match them is an empirical question that existing verified-env frameworks cannot pose.

We contribute: (i) a reward construction combining split-conformal coverage [Angelopoulos and Bates, 2023, Vovk et al., 2005, Lei and Wasserman, 2014] with task-specific point metrics, landing within Monte-Carlo error of the $1-\alpha$ target across all ten shipped envs; (ii) a procedural measurement-

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regeneration protocol that eliminates fixed-string memorisation as a valid strategy by construction; (iii) a benchmark of six frontier models across five scientific domains with Bonferroni-corrected paired-bootstrap significance tests; and (iv) ten open-source (MIT) environments on Prime Intellect Environments Hub [Prime Intellect, 2024].

2 Related Work

RLVR environments. Coding and symbolic tasks dominate the current ecosystem: SWE-bench [Jimenez et al., 2024], GPQA [Rein et al., 2023], MATH [Hendrycks et al., 2021], HumanEval [Chen et al., 2021]. Scientific inverse problems are absent. Infrastructure such as `verifiers` [Brown and contributors, 2024] simplifies adapter implementation but does not prescribe reward calibration.
Scientific ML. Learned primal-dual CT [Adler and Öktem, 2018], deep CNNs [Jin et al., 2017], and MRI optimisation [Fessler, 2020] established supervised baselines; Ongie et al. [2020] survey the space. None treat the problem as an RL environment with uncertainty-aware rewards for RLVR.
Conformal prediction. Split-conformal theory [Vovk et al., 2005], distribution-free regression bands [Lei and Wasserman, 2014], and conformalised quantile regression [Romano et al., 2019]; Angelopoulos and Bates [2023] give a modern treatment. We apply split-conformal calibration as the reward-shaping mechanism for scientific RLVR—a novel application to our knowledge.

3 Method

Environment framework. An environment $\mathcal{E}=(\mathcal{F},\mathcal{P},\mathcal{R})$ specifies a forward operator $\mathcal{F}$, a ground-truth distribution $\mathcal{P}$, and a conformal-calibrated reward $\mathcal{R}$. Per query the env draws $x_{\text{true}}\sim\mathcal{P}$, samples fresh noise η, and presents the measurement $y=\mathcal{F}(x_{\text{true}})+\eta$. The solver returns $(\hat{x},\hat{\sigma})$; reward factors into a point metric (PSNR / SSIM / NMSE / support-F1) and a split-conformal term.

Conformal-calibrated rewards. Fix $\alpha=0.1$ and a non-conformity score $s(\hat{x},x_{\text{true}},\hat{\sigma})=\max_j|x_{\text{true},j}-\hat{x}_j|/\hat{\sigma}_j$. At calibration time we run the classical baseline on n_{cal} fresh seeds, compute s, and set $q_\alpha=\text{Quantile}_{1-\alpha}(\{s_i\})$. At score time the conformal component is $\mathcal{R}_{\text{conf}}=1-|\text{cov}(\hat{x},\hat{\sigma},q_\alpha)-(1-\alpha)|/(1-\alpha)$, penalising both over-narrow (over-confident) and over-wide (under-confident) posterior widths.

Procedural regeneration. Each env factors into $(\text{seed},\text{image_id})$. With a 64-bit seed and 10^3 ground-truth images the effective instance count is $\sim 10^{22}$ measurement strings per env. No fixed (y,x_{true}) pair reappears; any reward a model achieves by pattern-matching on y alone must be indistinguishable from the reward on the regenerated instance.

Multi-turn and tool-use. Four envs expose a 3-turn dialogue where the server returns a forward-model residual between turns. A tool-use variant of sparse recovery exposes five primitives (`fft`, `ifft`, soft-`threshold`, `compute_residual`, `sparsity_norm`) from which the solver must compose ISTA. An earlier v0.1 design bundled an `ista_tool` oracle; empirical byte-identical rewards ($r=0.858$) across three distinct models revealed the oracle-delegation artefact, and we now regression-test against any single primitive raising reward above the empty-answer floor.

4 Experiments

Setup. Ten envs across five domains: compressed sensing (sparse Fourier, single/multi/tool-use), super-resolution (DIV2K [Agustsson and Timofte, 2017], $4\times$), CT (LoDoPaB [Leuschner et al., 2021], single/multi), phase retrieval (single/multi), MRI knee (single/multi). Baselines: OMP, bicubic, FBP, Gerchberg–Saxton, zero-filled IFFT. Models via OpenRouter: Claude Opus 4.7, Sonnet 4.6, Haiku 4.5, GPT-5.4, -mini, -nano [Anthropic, 2025, OpenAI, 2025]. Conformal: $\alpha=0.1$, $n_{\text{cal}}\geq 30$. Three instances per (env, model) cell. Total LLM API spend across all four paper-final benchmark CSVs: \$5.49. Paired-bootstrap significance testing uses 10k resamples.

Finding 1 – classical beats every tested LLM on most envs (Figure 2 L). Across 50 paired (env, model) comparisons and 10 000 paired-bootstrap resamples, classical significantly outperforms the LLM on 32 ($p<0.05$, both uncorrected and after Bonferroni correction for $m=50$); eleven show no significant difference, and the LLM significantly outperforms classical on seven (all at super-resolution, where the LLM is fed a bicubic-prior input and reward is bounded near the baseline). Pooled mean $\Delta=+0.199$ (classical $-$ LLM) with five-env cross-env means: classical 0.630; Haiku 4.5 0.558;

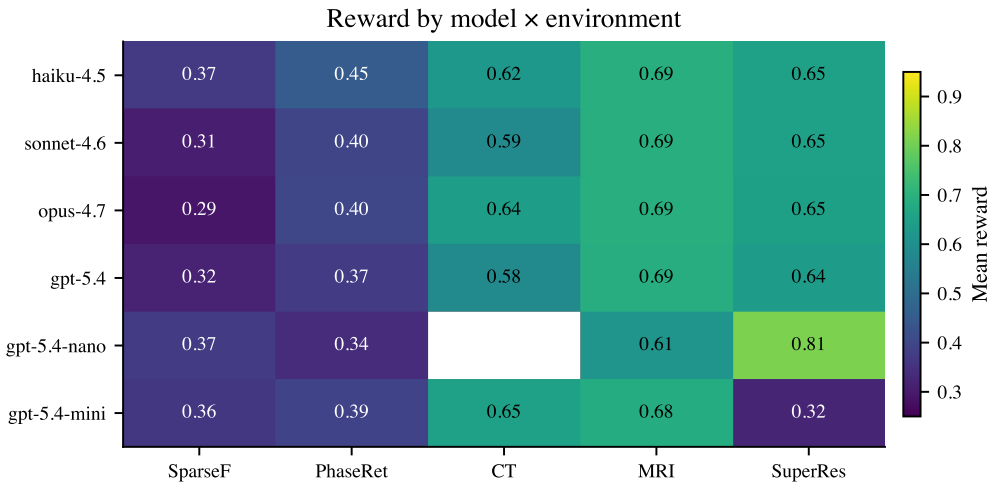

Figure 1: Reward per (model, env) across five single-turn envs. Six models, three instances per cell. Classical baselines printed below the chart; '—' denotes pairs with no parsed result.

GPT-5.4-nano 0.535 (4 envs only); Opus 4.7 0.534; Sonnet 4.6 0.527; GPT-5.4 0.519; GPT-5.4-mini 0.483. No LLM beats classical on a cross-env mean.

Finding 2 – multi-turn does not robustly help (Figure 3, appendix). Across the four multi-turn variants, only GPT-5.4 shows a meaningful positive Δ ($+0.029$ mean across 3 domains). Sonnet 4.6 is nearly flat ($+0.011$, 2 domains); Opus 4.7 (-0.012, 3 domains) and Haiku 4.5 (-0.036, 4 domains) regress slightly; budget models (GPT-5.4-mini -0.040, -nano -0.026) regress more. The earlier claim that multi-turn cleanly separates frontier from budget tiers (Sprint-1 v2) does not replicate with the complete matrix: multi-turn benefit is domain- and model-specific, not a simple tier effect.

Finding 3 – primitive composition gap (Figure 4, appendix). On the v0.3 primitive-only tool-use benchmark across all six models, per-model means are Opus 4.7 0.545 (n=1 parsed), Sonnet 4.6 0.482, GPT-5.4 0.459, Haiku 4.5 0.431, GPT-5.4-mini 0.401, GPT-5.4-nano 0.395 (n=1 parsed). Pooled mean 0.452, compared to classical OMP 0.870 on the same instances ($\Delta{=}+0.418$). For reference, the v0.1 oracle-tool artefact value (0.858) is shown in the figure.

Coverage validation (Figure 2 R). Empirical coverage of the conformal interval at $N{=}200$ fresh calibration samples per env. Grand mean $0.9013{\pm}0.0166$ against the 0.90 target; all per-env means in $[0.880, 0.931]$. The reward calibration delivers its distribution-free coverage guarantee empirically.

Cost efficiency. Per-episode LLM cost spans $36\times$ from nano ($\$0.0014$/ep) to Opus 4.7 ($\$0.0506$/ep), with Haiku 4.5 the cheapest in the top LLM cluster ($\$0.0095$/ep). Classical baselines cost zero LLM API. For a fixed RL training budget, cheap-tier LLMs generate $\sim 36\times$ more episodes, which matters when the training-signal gap is $0.072{-}0.199$ reward: cheap+many may beat expensive+few on wall-clock convergence. This is orthogonal to reasoning capability but central to practical RLVR deployment.

5 Discussion

Scientific inverse problems surface capability gaps that symbolic benchmarks do not: even the top-LLM 5-env mean (0.558) is 0.072 below the classical reference (0.630), and per-env bootstrap CIs separate classical from every LLM on 32 of 50 comparisons. The multi-turn result reshapes the narrative from Sprint-1: with complete data, residual-feedback gains are small and model-specific, not a clean tier separation. The primitive tool-use result is a negative capability finding—LLMs cannot, at the tested scales, execute classical compressed-sensing reasoning from primitives—that is itself useful RLVR training signal.

Limitations. Sample sizes are three seeds per (model, env) cell, so bootstrap CIs remain wide even when Bonferroni-corrected p-values are small. Images are rendered at $16{\times}16$ or $32{\times}32$ for LLM-token tractability; scaling to clinical resolutions is a v2 task. Phase retrieval is hard even for Gerchberg–Saxton ($r{=}0.29$), and GPT-5.4-nano had parse failures on lodopab-CT, so n=4 rather than

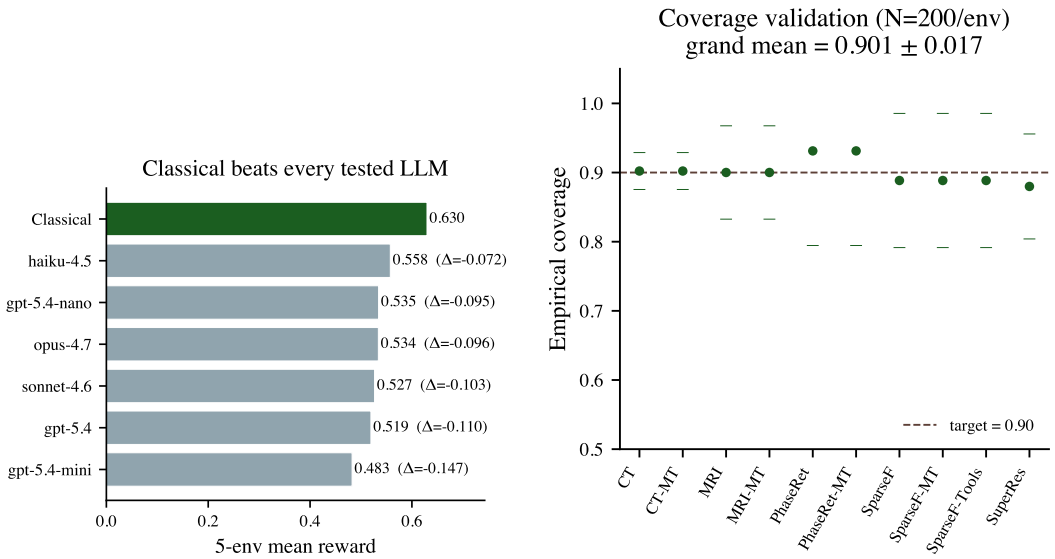

Figure 2: **(L)** Finding 1: 5-env mean reward per model vs classical baseline (0.630); top LLM is Haiku 4.5 at 0.558. **(R)** Conformal coverage at N=200 per env: grand mean 0.901±0.017 against the 0.90 target.

5 for its cross-env mean. Multi-turn budget enforcement aborted mid-run at 26% overshoot (reported honestly in artefacts), leaving MRI-MT and CT-MT incomplete for some models. We report no training experiments—only evaluation. **Future work.** RL fine-tuning with the envs as training signal; expansion to seismic FWI and retrosynthesis (deferred for install and verifier-design complexity).

6 Conclusion

Ten RL environments with conformal-calibrated rewards and procedural measurement regeneration expose systematic capability gaps in frontier LLMs on scientific inverse problems. Code: github.com/stelioszach03/verifiable-labs-envs; envs: app.primeintellect.ai/dashboard/environments/stelioszach; leaderboard: huggingface.co/spaces/stelioszach03/scientific-rl-benchmark.

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A Supplementary figures

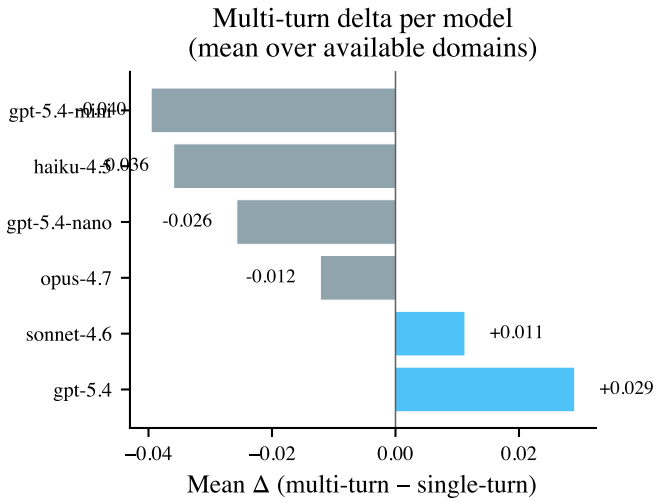

Figure 3: Mean reward delta (multi-turn minus single-turn) per model averaged over the multi-turn envs with paired data. GPT-5.4 is the only model with a meaningfully positive delta; all others are near-zero or negative. The clean frontier-vs-budget tier separation observed in Sprint-1 does not replicate on the complete matrix.

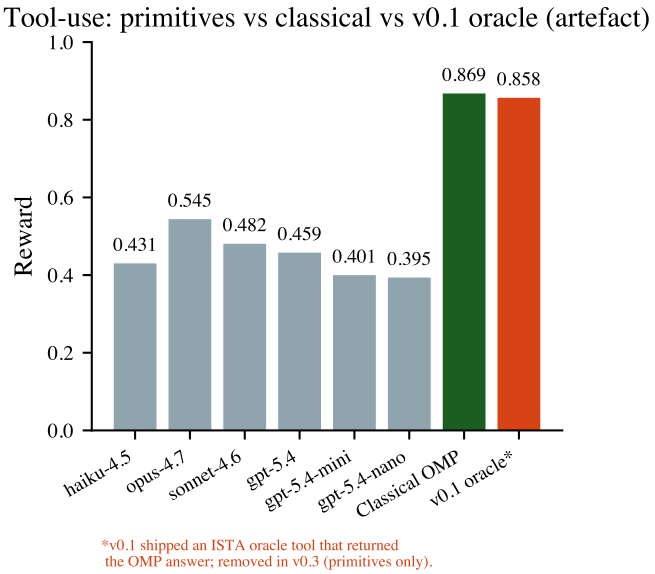

Figure 4: Tool-use rewards on sparse Fourier recovery (v0.3 primitive tools only). Six LLMs cluster in $[0.40, 0.55]$; classical OMP on the same instances reaches 0.87. The v0.1 oracle-artefact value (0.858) is shown for contrast: it matched classical precisely because the tool itself returned the classical answer.

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